ALIoT Smart Lab - engineering dossier

Version 2026-10-20 **evening update** (the morning 2026-10-20 and the 2026-10-18 dossiers stay as history). Branch `offline-router-door-photos`, committed as `b8c5bdb`, **not pushed**. Deployed portal version `f5eaeff-dirty-20261020021246` (nothing redeployed since - only CAD and docs changed). Database migration head `b7c3e1f9a204`. Labels: LIVE-VERIFIED seen working on the real system, TESTED OFFLINE automated tests only, SIMULATED, PHYSICAL VALIDATION REQUIRED, BLOCKED / OWNER INPUT.

New in this update (section 0): the master-door enclosure fit-test release (datasheet research, SOLIDWORKS model, interference check, drawings), the Face-ID photo self-service, the camera-first GPU gate, the FANUC / UR5e welcome view, the AI CPU-vs-GPU measurement and the evening test run. The speaker / PAM8403 was removed by the owner and is gone from the wiring record.

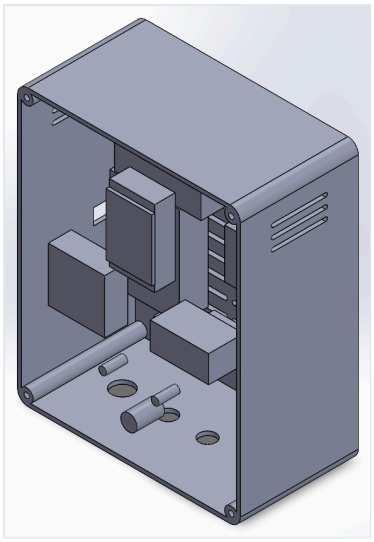
0. This update (2026-10-20 afternoon / evening)

Area	What was done	Status
Door node enclosure	Every opening re-based on sourced dimensions (4 research passes over Most Electronics, Ekostra and the manufacturers' datasheets; report `reports/Door enclosure component dimensions.md`). New mounts: TFT on its 4 PCB holes, PN532 on islands behind a 1.5 mm skin, R307 in a cradle with a cable tie (it has no flange), active-buzzer cradle with sound holes, PG7 / M16 gland holes. The 180 x 150 mm fit coupon tests every one.	VERIFIED IN CAD FIT-TEST PRINT REQUIRED
Virtual assembly	Body + front panel + plate + 14 component envelopes (outline + clearance). SOLIDWORKS interference detection: **no interferences (17 bodies)**. The first run found the ESP32 overlapping a lever connector by 0.25 mm and a square buzzer envelope inside its round cradle - both fixed. 25 mm wire space between the plate modules and the front modules.	VERIFIED IN CAD
Printable files	Body, front panel, plate, coupon: STL (all closed, manifold, consistent winding, no degenerate triangles - `check_stl.py`), STEP, SLDPRT; assembly SLDASM + STEP; dimensioned drawings PDF; release ZIP.	VERIFIED
Labels	Title "DOOR NODE (MASTER ESP)" and placement labels engraved 0.6 mm, panel printed front-down (smoothest face; raised text is printed next to it on the coupon for comparison). Bug found and fixed: SOLIDWORKS sketch text is left-aligned, so the title ran off the panel edge - every text is now measured and centred.	VERIFIED IN CAD
Face-ID photo self-service	Users send photos (file or selfie); the portal checks them with the face server (no face / someone else's face refused); staff approve (trains), ask to retake, or refuse ("see the lab staff"); notifications both ways.	LIVE deployed; 16 backend tests
Camera-first GPU gate	One decision for both assistants: any camera live or unknown -> CPU at once; GPU only after 120 s idle, at most once per 300 s, with free GPU memory. Opt-in (default unchanged).	TESTED OFFLINE 9 + 5 tests
Welcome view	FANUC cell turned so its real front faces the viewer; UR5e starts at an easier home pose.	LIVE

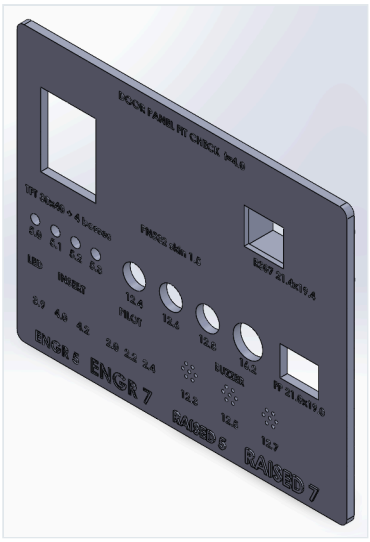
Enclosure - renders and drawings



front panel



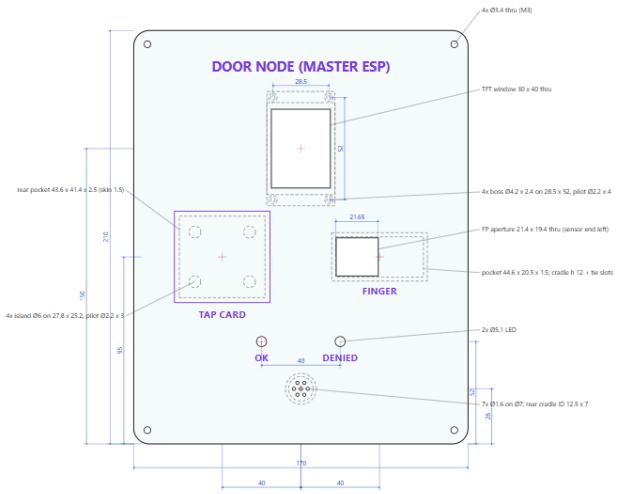
inside (panel hidden) with component envelopes



fit coupon - print first

Front panel - front view

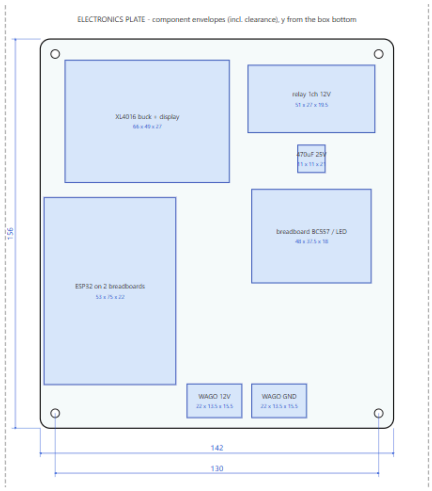
seen from the corridor; dashed = rear features (pockets, bosses, cradles); purple = engraved



ALIoT Smart Lab - MASTER_LAB01 door node enclosure source: tools/cad/door_enclosure/params.json + Build.cs (SOLIDWORKS 2025)	units mm general tol. ±0.3 (FDM)	scale ≈1:1.8 2026-10-20	sheet 1 / 4 status: FIT-TEST RELEASE

Electronics plate - layout

component envelopes as used in the SOLIDWORKS interference check



ALIoT Smart Lab - MASTER_LAB01 door node enclosure source: tools/cad/door_enclosure/params.json + Build.cs (SOLIDWORKS 2025)	units mm general tol. ±0.3 (FDM)	scale ≈1:1.3 2026-10-20	sheet 3 / 4 status: FIT-TEST RELEASE

Enclosure - where each opening comes from

Part (seller)	Key dimensions used	Confidence
R307 fingerprint (Most ME-30297)	44.1 x 20 x 23.5, head face 19 x 21, sensing window 15 x 11.19, no flange (GROW manual p.2)	DATASHEET (no tolerances published)
1.8" ST7735 TFT, red V1.1 (Most ME-38699)	PCB 34.5 x 58, holes D3.2 on 28.5 x 52 (LCDWIKI MSP1803); glass 34.7 x 46.7 x 2.4, active area 28.03 x 35.04 (QDTFT1801)	VARIANT + DATASHEET
PN532 V3 red (Ekostra EK506)	42.7 x 40.4 x 4; 2 holes D3.05 on 27.8 x 25.2 (Elechouse V4 STEP, "same holes as V3")	VARIANT / REFERENCE (holes)
Active buzzer (Most ME-32997 / ME-19495)	D12 x 9.5 (listing); TMB12A05 D12.0 +-0.2	VARIANT / DATASHEET

ESP32 30-pin USB-C, XL4016 + display, PCT-212 lever connector (all Most)	51.5 x 28.5; ~64 x 47 x 26; 21 x 12.5 x 14.5 - no hole patterns published, so the plate has slots everywhere	VARIANT (outline only)
Cable glands (not sold by Most), M3 inserts	PG7 hole 12.5-12.8, M16 16.0-16.3; insert hole 4.0, min. wall 1.6 (CNC Kitchen)	DATASHEET

Full register: `docs/hardware/DOOR_ENCLOSURE.md`. Verdict: **ready for the fit-test print, not yet for the final box** - most hole patterns are vendor / reference data, not measured on the owner's parts; the coupon checks them all in one ~97 cm³ print.

AI assistants: CPU vs GPU, cameras idle MEASURED

qwen2.5:1.5b, 120-token answers, n = 6 each	Answer (median)	Generation	GPU memory
CPU (current setting)	2.4 s	22 tokens/s	0 MB
GPU	1.3 s	68 tokens/s	1062 MB
CPU again (left as found)	2.6 s	20 tokens/s	0 MB

Both camera services were up with no stream (0 analyses/s). Short answers are quick on the CPU too; the 51 s CPU figure in section 2 is a long answer with document context. The GPU generates ~3x faster - worth it only while the cameras are idle, which is what the opt-in GPU gate does. `tools/benchmarks/ai_cpu_gpu_ab.py`.

> **State while the morning sections (1-7) were written:** the laptop was off the lab network (phone hotspot). Both camera services run the new code but have no stream; no ESP node, robot or camera could be tested live after 19:15 on 2026-10-19. Portal changes were deployed and verified on the live portal; tracking changes are tested offline only.

1. What changed in this round

Area	Root cause (evidence)	Fix	Status
USER1 / USER5 cross-camera conflict	Camera 1's desk rule named whoever sat at PC-09/PC-10 "USER5" from a door entry hours old; camera 2's face said USER1. 4 episodes (16:14, 16:15, 17:02, 18:36), 243 conflict lines, 546 of 547 USER5 namings came from the desk.	The desk only continues evidence: name on a track within 120 s, or door / face / other camera within 10 min. Otherwise `DESK_NOT_USED`.	TESTED OFFLINE regression test fails on the old code
Seated-person flicker	The labeler saw only tracks detected in the last 0.6 s while the tracker keeps a still track 6 s: every short miss logged NAME_LOST and let the desk name a second track (NAME_ON_TWO_PEOPLE x24).	Tracks still kept by the tracker ("coasting") keep their names and are not reported lost; hand-back counts from last-seen time.	TESTED OFFLINE
Nour Unknown after a short track break	Hand-back looked only at the spot where the track was lost; a walking person reappears further along.	Search along the walking path (look must still agree).	LIVE-VERIFIED 19:05 walk test

Wrong name after a crossing (camera 2, 19:05)	Nour's name, correctly handed over, stayed on camera 2's track after it went onto Ali; camera 2's own face said USER1; by design the name was not switched.	A name only carried over (other camera, desk) gives way to this camera's 2 agreeing face matches when nobody touches. Door names keep the mismatch flag. Face thresholds unchanged.	TESTED OFFLINE
FANUC Educational Cell	Drawn from overall dimensions only; panels 12 % opaque; yellow panel on the wrong face; no lettering; cell front opposite the owner's real front.	Glass with edges, two doors, hinges, knob, FANUC sticker, yellow side panels ("Learn online: fanuc.eu/educational-package"), controller with e-stop, pendant, crate, slotted table, parts plate, "FANUC Robot ER-4iA" on the forearm; robot right of the plate, front = owner's front.	LIVE-VERIFIED deployed, screenshots
Robot mesh size	UR5e meshes were unindexed (each position stored ~5.6 times); the 2026-10-19 "12.5 MB -> 75 KB" counted only the index files.	Weld + quantize, every triangle kept: UR5e + Robotiq 10.4 -> 1.9 MB, FANUC 1.26 -> 0.26 MB.	LIVE-VERIFIED renders identical
System tab slow	12 Recharts charts built at once (643 ms block) plus entry animations (~100 ms blocks repeating).	Charts built when near the screen; animations off on that page.	LIVE-VERIFIED
Evacuation alarm missing on full-screen pages	Alarm and emergency message mounted only in the portal frame - /welcome (every sign-in lands there), /monitor, /wall, /report showed neither.	Safety layer on those pages; Welcome gets live updates.	TESTED OFFLINE e2e on the isolated copy; deployed
Stale data after a write	Identical GETs shared one answer for 1 s, also across a write: a photo uploaded within 1 s of opening an issue stayed invisible.	Every write clears the shared reads.	TESTED OFFLINE workflows 30/30; deployed
Welcome finale crash (own regression)	The mesh-loader change sent Kinova `.STL` files through the glTF loader (deployed 23:36).	Only the converted FANUC uses .glb; fixed and redeployed 00:01.	LIVE-VERIFIED welcome 20/20 admin+staff

2. Measurements

Portal navigation (deployed build, real browser, admin)

Page	Before (first / repeat)	After (first / repeat)
Robots & PCs tab (cold, empty cache)	2.0 s, 8.3 MB models	1.5 s, ~1 MB models
Robots & PCs tab (repeat visit)	0.78 s	0.29 s
System analytics	1.20 / 0.88 s	0.27 / 0.23 s
Reliability	0.96 / 0.84 s	0.08 / 0.04 s
All other sidebar tabs	66-693 ms	40-500 ms

Medians of 3 runs (robots) / 2 rounds (tabs). Scripts: `frontend/e2e/robots_tab_perf.mjs`, `nav_timing.mjs`. Remaining cold cost is main-thread work (3D library start, WebGL), not downloads.

Load test SIMULATED (isolated backend + test DB, 120 s)

Measure	Result
API errors	0
API p95	all < 220 ms; `/summary` one sample of 12 at 525 ms; `/monitor` p95 215 ms
Two simulated cameras (GPU, DirectML)	6.5 / 6.9 analyses/s, frame age p95 ~85 ms, GPU 50 %, VRAM 1155 MB
Memory / threads	RSS 2163 -> 2171 MB, threads 300 -> 283 (no growth)
WebSocket per client	robot_status 1.1/s, env 0.36/s, security 0.19/s

AI assistants LIVE-VERIFIED

Structured answers 0.0-0.2 s; document lookup 1.6 s; an answer from the language model (qwen2.5:1.5b on the **CPU**, cameras idle) 51.5 s. CPU is the owner's choice (GPU for the cameras; GPU answers were ~3 s on 2026-10-19).

Cross-camera evidence (camera logs 2026-10-19)

Measure	Value
Appearance distance, same person (both names confirmed)	n = 119, median 0.30, p90 0.44, max 0.73
Appearance distance, different people	n = 2 (0.67, 0.70) - the earlier 0.28 / 0.38 pairs used the wrong desk name and are excluded
Floor gap, same person in both cameras	median 0.41 m, p90 0.86 m (pairing gate 0.7 m)
Floor coverage from the two floor maps	both cameras: middle of the room (y ~3.5-8 m); camera 1 only: far end; camera 2 only: near end; neither ~1 % (occlusion not modelled)
Handover at entry (19:04 walk test)	camera 1 named at the door, camera 2 had the name 4 s later; longest Unknown on camera 2: 35 s after a track break where only camera 2 saw the person
Simulated scenario bench (14 scenarios)	no false names, no name switches - identical to 2026-10-18 SIMULATED

Conclusion: appearance matching stays observe-only - not because it is shown unreliable, but because only 2 different-person pairs exist. The blind-spot Unknown time is track breaks in camera 2's own area, not unseen floor.

3. Tests

Suite	Result
Backend (pytest, smartlab_test)	**554 passed** (8 min 18 s, evening run; 542 in the morning)
Security cameras	253 passed (11 new regression tests; evening re-run 253)
Robot monitor / knowledge / PC agent	41 / **49** / 5 passed (evening run)

Frontend	TypeScript clean; production build OK (evening re-run)
Enclosure CAD	interference check: none (17 bodies); STL check 4/4 closed + manifold
Browser e2e, isolated seeded copy	workflows 30/30, welcome 27/27, reliability 31/31, monitor+security 18/18, emergency message 6/6, notifications, experiments, report link pass; navigation 97 checks, 2 console messages = camera 503 (test cameras stopped)
Browser e2e, live portal (read-only)	robot readiness 8/8, monitor+security 18/18, reliability 31/31, welcome 20/20 (admin, staff)
Not run	camera recording e2e (cameras offline); student-role checks on live (real account, not the demo password)

Six e2e scripts were updated for UI changes made since they were written (sign-in now lands on /welcome; booking wizard has an Equipment step; lab count read from the database).

4. Devices

Device	State
Door master, environment node, consumables bin, reusable bin, PC agent	last report 2026-10-19 03:59-04:29 local; **none reported during the 2026-10-19 lab session** although the cameras on the same network did PHYSICAL CHECK
ESP32-CAM + face server	face server up (SFace, threshold 0.363); templates USER1, 2, 3, 5, 6, 7 - **Hemaly (USER5) is loaded** (23 samples); live door test needs her present PHYSICAL
Robots (UR5e, FANUC)	collectors running read-only, no data off the lab network; DRY RUN, nothing sent
Cameras	services healthy, reconnecting; 7 fps each when live (2026-10-19)

5. Remaining - owner actions

#	Item	Type
1	Walk tests at the lab (one person, then two crossing) to confirm the tracking fixes live; record a 2-person session for appearance thresholds	PHYSICAL
2	Only if the walk shows long camera-2-only stretches: tilt camera 2 and re-click the tape marks (back up the floor JSON first)	PHYSICAL
3	PC letter -> number mapping for camera 2 (B, C, D, F, G, I, J, K, L -> PC-01..11)	OWNER INPUT
4	Hemaly at the door: positive + negative face test	PHYSICAL
5	Why the ESP nodes stopped reporting at 04:00 (power, Wi-Fi, portal address)	PHYSICAL
6	**Print the enclosure fit coupon** (same material as the panel, front down) and try the real TFT, PN532 (a card must read through 1.5 mm), R307, LEDs, buzzer, inserts, glands; measure the R307 length (44.1 or ~52 mm) and the breadboard height (8.5 or 10 mm)	PHYSICAL
7	Buy 3 x PG7 + 1 x M16 cable glands with locknuts (Most Electronics has none)	OWNER
8	Door wiring values not recorded (section 6) - measure	PHYSICAL
9	Full 3-run before/after benchmark of every sidebar entry: needs `E2E_PASSWORD` set in the environment (the tooling refuses a password typed on the command line)	OWNER

10	Decide whether to switch the assistants to the GPU gate (auto) - numbers in section 0	OWNER

6. Appendix - door master (MASTER_LAB01) wiring as recorded

From the firmware constants and the 2026-10-04/05 bring-up log; full version with parts table: `docs/hardware/DOOR_MASTER_WIRING.md`. "Not recorded" = measure before use; nothing was guessed.

ESP32 GPIO	Connected to	Notes
19 / 18	PN532 V3 (HW-147) SDA / SCL	I2C 0x24, S1 ON / S2 OFF, IRQ + RESET not wired; supply pin not recorded
13 / 2	ST7735 1.8" TFT SCK / MOSI	own SPI bus; GPIO2 is a boot strapping pin
25 / 32 / 33	ST7735 CS / DC / RST	VCC and backlight supply not recorded
16 / 17	fingerprint TX -> ESP RX16; ESP TX17 -> sensor RX	AS608-protocol, 57600 8N1; model and supply not recorded
27	MC-38 reed switch to GND	INPUT_PULLUP: LOW closed, HIGH open
26	BC557 base via 3.3 kΩ	HIGH = locked, LOW = unlocked
14	green LED	resistor not recorded
21	NPN stage: red LED + buzzer	transistor and resistors not recorded; 600 ms denial beep
15	red LED via 220 Ω to GND	own pin, no buzzer current
22	nothing now (was 3.3 kΩ -> PAM8403)	**speaker + PAM8403 removed** (owner, 2026-10-20) - the tablet page plays the sounds

```
+12 V -- relay VCC, relay COM, 470 uF 25 V (+) ;  (-) -- GND
relay NO -- solenoid (+) ;  solenoid (-) -- GND            (lock on NO = fail-secure)
3V3 -- BC557 E ;  BC557 C -- relay IN ;  BC557 B -- 3.3 k -- GPIO26 ;  15 k between B and E
GPIO26 HIGH -> PNP off -> coil off -> locked.   LOW -> ~1 mA base -> coil on -> unlocked.
Measured: COM-GND 12.1 V at rest, 12.03 -> 11.95 V during an unlock.
Supply: 12 V / 10 A PSU -> master switch -> WAGO -> fused branch -> buck -> 5 V -> ESP32 VIN (door
buck model not recorded).
```

Records disagree: `door_audio.h` says RC ~47 µs, but 3.3 kΩ x 100 nF = 330 µs; `docs/DOOR_SYSTEM.md` pin table still shows the retired MFRC522 and the lock on NC.

7. Evidence files

`docs/camera-identity/TWO_CAMERAS.md` (2026-10-20 section), `docs/PROGRESS_CHECKPOINT.md`, `docs/hardware/DOOR_MASTER_WIRING.md`, `tools/security_vision/test_desk_evidence_2026_10_19.py`, `test_face_overrides_carried_name.py`, `test_walk_handback.py`, `frontend/scripts/compact_glb.mjs`, `stl_to_glb.mjs`, `frontend/e2e/robots_tab_perf.mjs`, `backups/deploy_history.jsonl`; enclosure: `tools/cad/door_enclosure/` (params.json, Build.cs, out/), `docs/hardware/DOOR_ENCLOSURE.md`, `reports/Door enclosure component dimensions.md`; AI: `docs/reports/ai_cpu_gpu_ab_20261020.json`. Camera logs (git-ignored): `tools/security_vision/recordings/identity_log_*.jsonl`, `crosscam_obs_*.jsonl`.